Verification Guide

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Abstract

This guide gives reproducible commands and the exact scope of the 15-stage finite replay for the main manuscript. The publication verifier checks file integrity and source bindings. The scientific replay recomputes the finite inequalities in separate consumers. Both are required to assess the computational part of the package; the written analytic arguments and external theorem input retain their stated review boundaries.

1 Edition and contents

This edition supersedes referee edition 3 of 28 October 2026; the numerical plan, source table and comparison history are unchanged; Appendix B adds the direct source reconstruction. The fifteen-stage replay accompanies DOI 10.5281/zenodo.23752103; the release receipt records the final manuscript page count.

The public set contains 16 files: the main PDF and TeX, three current supporting PDFs, two preserved prior PDFs, manuscript sources, the proof archive, README, citation and rights files, release manifest, execution receipt, publication verifier and checksums. The archive retains exact raw inputs, the full runtime and corresponding source distributions, the 67 kernel certificates, the full 221,010-arc history, and attributed external theorem evidence.

2 Check the delivered files

Place all 16 public files in one directory. Python 3.12 is the tested interpreter. The publication-integrity checker uses only the standard library.

```
python3.12 Verify_Deposit.py
```

This checks the exact public inventory, SHA-256 digests, nested archive inventories, source/PDF bindings, source identity, recorded stage list and exact endpoint arithmetic. It does not rerun the mathematical consumers. A PASS from this command must not be described as a new mathematical replay.

3 Extract and run the scientific replay

Extract `Proof_and_Replay.zip`. From the directory containing its `Proof_and_Replay/` folder, run:

```
cd Proof_and_Replay/Lambda_Peer_Review
python3.12 verify_bundle.py --all --jobs 4 \
  --work ../../lambda-replay-fresh
```

The work directory must be outside the immutable scientific bundle. Choose a new directory for a fresh run. Do not use Python's `-O` or `-OO` options: assertions must remain enabled. The launcher maps the retained historical path strings to the bundled input tree. The ordinary replay has no historical filesystem fallback and needs no external research workspace.

The included runtime is the retained python-flint 0.8.0 Linux CPython 3.12 runtime with FLINT 3.3.1. It is not a cross-platform binary. A reviewer on another platform can install a compatible python-flint environment and supply its site-packages directory using the `-runtime` option. The numerical implementation and version must be recorded with the new execution receipt.

To list the actual stages, or execute one stage after its dependencies have produced the required work files:

```
python3.12 replay/run.py --list
python3.12 replay/run.py --step direct-source \
  --work ../../lambda-replay-fresh
```

Running a single stage does not establish a complete fresh replay.

4 Dependency order and acceptance scope

Stages in a dependency group may run in parallel. A group completes successfully before the next group begins.

Stage	Obligation
Group 1	
`native`	All 67 rows and 469 native heights; primitive finite prefix and tail enclosures.
`activation`	Both native branches over the entire activation half-line.
`spectral`	Continuous spectral envelopes of every signed scalar filter.
`heads28`	Complete shared-row coefficient capacities, fixed allocation, coordinate reserve and weighted Gram payment.
`heads39`	Complete retained-row heads, prime/composite resources and original joint angular payment.
`wall`	Continuous wall and complete physical transfer.
`kernel`	All 67 continuous kernel certificates and whole signed own-pair bounds.
`arcs`	Every one of the 221,010 exact comparison arcs, startup, joins and terminal cleanup.
Group 2	
`affine`	Retained-row native intercepts, near/far joins and remaining main slopes.
`overlays`	Physical refunds, joint derivative source bounds and total replacement common-clock gains.
`compact-arcs`	Compact exact all-arc margin transcript derived from the fresh arc witnesses.

Stage	Obligation
Group 3	
`interfaces`	Exact functional, thermal/contact and source-field identities between the fresh components.
`physical-debit-map`	Explicit composition of the physical debit ledger.
`arithmetic-credit-map`	Explicit composition and accounting of the arithmetic credit ledger.
Group 4	
`direct-source`	New direct primitive reconstruction, actual main bounds, full jet binding and all 268 final-field identities.

The new direct stage makes formerly implicit bindings explicit. It does not replace the native, activation, spectral, head, clock or analytic proofs with a comparison of final numbers. No saved historical success flag is a mathematical acceptance premise.

5 The supplied fresh execution

The package includes the actual per-stage logs, access records, runtime identification and execution receipt of the new run. All 15 registered stages returned exit code zero. The source and input hashes bind that run to this reconstructed edition. Timing measures an execution and has no mathematical role.

The larger all-arc rational witness stream is a reproducible intermediate; the complete flow inputs, exact consumer and compact margin transcript are included. A new full run regenerates its working outputs. Historical 14-stage receipts are retained under the baseline provenance and are not presented as evidence for the new fifteenth stage.

6 Review the theorem interfaces

The initial theorem is OpenAI's exported ordinary Riemann-zeta statement at the pinned revision, with attributed external build evidence. This work did not perform a fresh full local Lean build. The heat-flow comparison is not formalised end to end in Lean.

The source identity requires one physical H_t, its actual central cutoff, translated points and normaliser. The kernel identity requires the complete signed scalar and derivative functional and the same contact cap. The global argument requires the full finite head, compact positive-time contact, repeated-contact exclusion and all closed time joins. Those are mathematical review tasks in the manuscript, even when every finite consumer passes.

The terminal value is

$$\begin{aligned} B_* &= \frac{22101}{204800} + \frac{1}{2}\frac{7902377963830536747158643}{2^{96}} \\ &= \frac{427693072077942473661578098491}{3961408125713216879677197516800} \\ &< 0.107964910079. \end{aligned}$$

The positive terminal squared height must receive its complete half-width contraction time. No conclusion $\Lambda < 0.1$, RH, Lindelof, optimality or external journal acceptance is asserted.

7 Build the documents

Extract `Manuscript_Sources.zip`. From its extracted source folder, use the supplied portable builder with a separate output directory:

```
python3.12 BUILD_PDFS.py --output ../document-build
```

The build requires pdfLaTeX and the packages named in the four TeX preambles. Sources are self-contained; no external figures are fetched. The builder preserves inputs, writes logs and source/PDF digests, and checks references and overfull boxes. PDF creation metadata can change the byte hash on another system; the source identity and rendered content are the appropriate separate checks.